4.6.2. A.C. through pure resistive circuit.

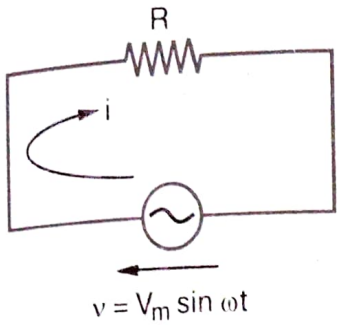


Fig. 4.14. Resistive Circuit

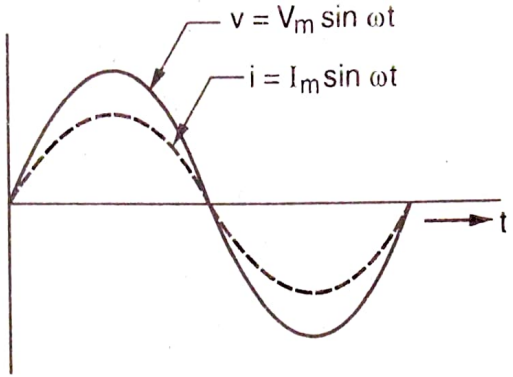


Fig. 4.15. Current and voltage waveforms.

Applied voltage, $v = V_m \sin \omega t.$ ...(4.37)

The resultant current, i

$$i = \frac{V}{R} = \frac{V_m}{R} \sin \omega t = I_m \sin \omega t \quad \text{...(4.38)}$$

Current, i is in phase with applied voltage, v. The waveforms and phasor diagram are shown in Fig. 4.15 and Fig. 4.16 respectively.



Fig. 4.16.

Power factor . Cosine of the angle between voltage and current

$$\text{p.f.} = \cos \phi$$

$$\phi = 0$$

So, $$\text{p.f.} = \cos 0 = 1$$

Power. Instantaneous power,

$$p = vi = V_m I_m \sin^2 \omega t$$

$$= \frac{V_m I_m}{2}(1 - \cos 2\omega t) = \frac{V_m I_m}{2} - \frac{V_m I_m}{2} \cos 2\,\omega t \qquad ...(4.39)$$

Power consists of a constant part $\frac{V_m I_m}{2}$ and a fluctuating part $\frac{V_m I_m}{2} \cos 2\omega t$ of frequency double that of voltage and current waves. For a complete cycle, the average value of $\frac{V_m I_m}{2} \cos 2\omega t$ is zero.

Hence, power for the whole cycle is

$$P = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} \qquad ...(4.40)$$

or $$P = V \times I \text{ watt}$$

where V = r.m.s. value of applied voltage,

I = r.m.s. value of the current

Power factor $\cos \phi$,

$$\phi = 0^\circ$$

So, $\cos \phi = 1$ (unity),

4.6.3. A.C. through pure inductive circuit. The circuit is shown in Fig. 4.18. Again the applied voltage is given by

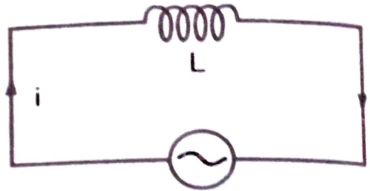


Fig. 4.18. Circuit diagram.

$$v = V_m \sin \omega t$$

$$v = L \frac{di}{dt}$$

But the voltage across the inductor is,

$$\therefore \quad V_m \sin \omega t = L \frac{di}{dt}$$

$$\therefore \quad di = \frac{V_m}{L} \sin \omega t \, dt$$

Integrating both sides, we get,

$$i = \frac{V_m}{L} \int \sin \omega t \, dt$$

$$= \frac{V_m}{\omega L} (-\cos \omega t) = -\frac{V_m}{\omega L} \cos \omega t \qquad \text{...(constant of integration = 0)}$$

$$\therefore \quad i = \frac{V_m}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right) = \frac{V_m}{X_L} \sin\left(\omega t - \frac{\pi}{2}\right) \qquad \text{...(4.41)}$$

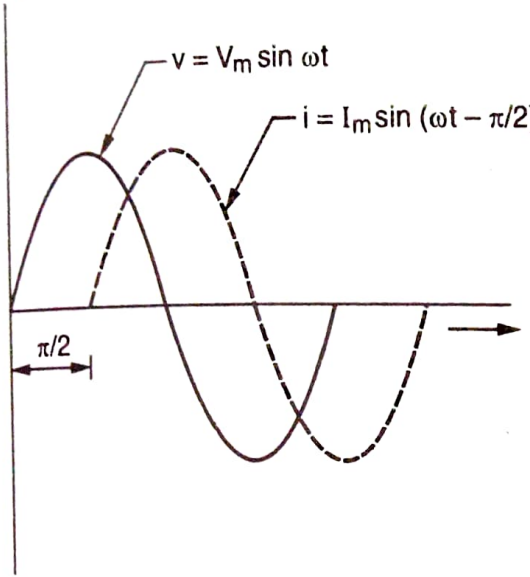
Maximum value of i is $I_m = \frac{V_m}{\omega L}$ when $\sin\left(\omega t - \frac{\pi}{2}\right)$ is unity.

Hence, the equation of the current becomes

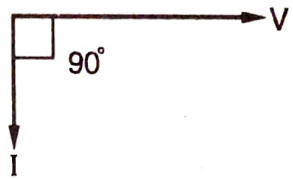
$$i = I_m \sin\left(\omega t - \frac{\pi}{2}\right) \qquad \text{...(4.42)}$$

So, we find that if applied voltage is represented by $v = V_m \sin \omega t$, then current flowing in a purely inductive circuit is given by $i = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$.

Clearly, the current lags behind the applied voltage by a quarter cycle [Fig. 4.19 (*a*)] or the phase difference between the two is $\pi/2$ with voltage leading. Vectors are shown in Fig. 4.19 (*b*) where voltage has been taken along the reference axis. We



(a) Voltage Current Waveform



(b) Phasor Diagram

Fig. 4.19.

have seen that $I_m = V_m/\omega L = V_m/X_L$. Here '$\omega L$' plays the role of 'resistance'. It is called the (inductive) *reactance* X_L of the coil and is given in ohms, if L is in henry and ω is in radian/second.

Now, $X_L = \omega L = 2\pi f L$ ohm. It is seen that X_L depends directly on frequency of the voltage. Higher the value of f, greater the reactance offered and *vice-versa.*

Power. Instantaneous power

$$= vi = V_m I_m \sin \omega t \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$= -V_m I_m \sin \omega t . \cos \omega t = -\frac{V_m I_m}{2} \sin 2\omega t \qquad ...(4.43)$$

Power for whole cycle is $P = -\frac{V_m I_m}{2} \int_0^{2\pi} \sin 2\omega t \, dt = 0$ ...(4.44)

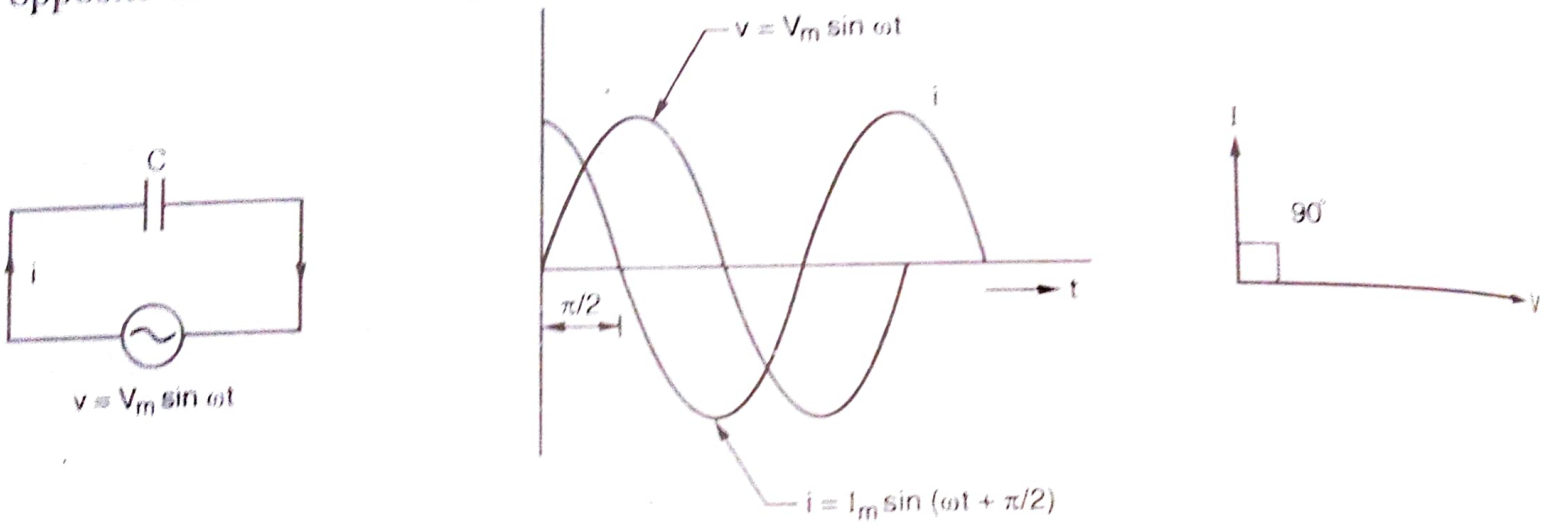
So the average power taken by the inductor for the complete cycle is zero.

Power factor. Angle between voltage and current,

$$\phi = 90^\circ \text{ (lagging)}$$

So $$\text{p.f.} \cos \phi = \cos 90^\circ = 0 \text{ lagging.} \qquad ...(4.45)$$

4.6.4. AC through pure capacitor. When an alternating voltage is applied to the plates of a capacitor, the capacitor is charged first in one direction and then in the opposite direction. See Fig. 4.20 (*a*).



(a) Circuit diagram (b) Voltage and Current waveform (c) Phasor diagram

Fig. 4.20.

v = p.d. developed between plates at any instant

q = Charge on plates at that instant.

The $q = Cv$ ... where C is the capacitance

Putting the value of v.

$$= CV_m \sin \omega t$$

Now, current i is given by the rate of flow of charge.

$$\therefore \quad i = \frac{dq}{dt} = \frac{d}{dt}(CV_m \sin \omega t) = \omega CV_m \cos \omega t$$

or $$i = \frac{V_m}{1/\omega C} \cos \omega t = \frac{V_m}{1/\omega C} \sin\left(\omega t + \frac{\pi}{2}\right)$$

Obviously, $$I_m = \frac{V_m}{1/\omega C} = \frac{V_m}{X_c}$$

$\therefore$ $$i = I_m \sin\left(\omega t + \frac{\pi}{2}\right) \qquad ...(4.46)$$

The denominator $X_C = 1/\omega C$ is *known as capacitive reactance* and is in ohms, if C is in farad and ω in radian/second. It is seen that if the applied voltage is given by, $v = V_m \sin \omega t$, then the current is given by, $i = I_m \sin (\omega t + \pi/2)$.

Hence, we find that the current in a pure capacitor leads its voltage by a quarter cycle as shown in Fig. 4.20 (*b*) or phase difference between its voltage and current is $\pi/2$ with the current leading. Vector representation is given in Fig. 4.20 (*c*). Note that V_c is taken along the reference axis.

Power. Instantaneous power,

$$p = vi = V_m \sin \omega t . I_m \sin (\omega t + 90^\circ)$$

$$= V_m I_m \sin \omega t \cos \omega t = \frac{1}{2} V_m I_m \sin 2\omega t \qquad ...(4.47)$$

Power for the whole cycle $= \frac{1}{2} V_m I_m \int_0^{2\pi} \sin 2\omega t \, dt = 0$.

We find that in a purely capacitive circuit, the average demand of power from supply is zero (as in a purely inductive circuit).

Power factor. Angle between voltage and current,

$$\phi = 90^\circ \text{ (leading)}$$

So $$\text{p.f.} = \cos \phi = \cos 90^\circ = 0 \text{ (leading)}.$$

So

4.6.5. AC through resistive-inductive circuit. A pure resistance R and a pure inductive coil of inductance L are shown connected in series in Fig. 4.21.

Let, V = r.m.s. value of the applied voltage,

I = r.m.s. value of the resultant current,

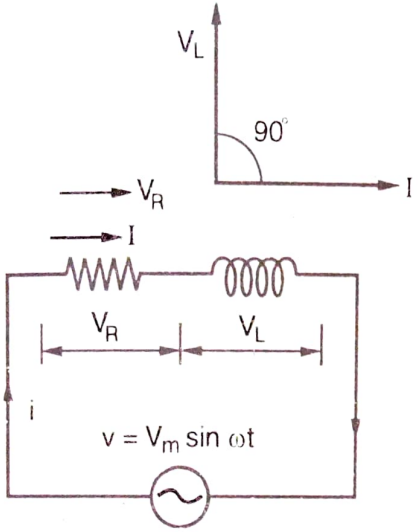


Fig. 4.21

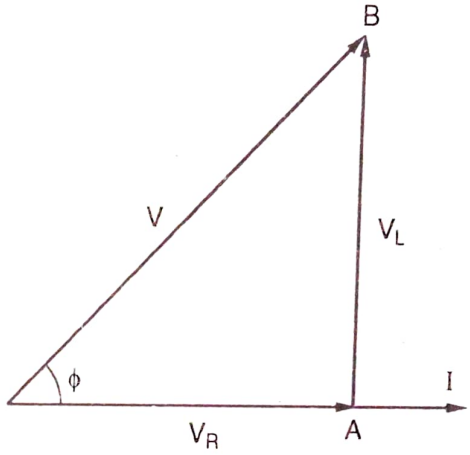


Fig. 4.22

$V_R = IR$ = voltage drop across R (in phase with I),

$V_L = I.\ X_L$ = voltage drop across coil (ahead of I by 90°)

Total impedance of the circuit is given by

$$Z = R + j\,X_L = R + j\,\omega L \qquad \ldots(4.48)$$

In polar form, $$Z = \sqrt{R^2 + X_L^2}\ \angle \tan^{-1}\frac{X_L}{R}$$

The resultant current is given by

$$i = \frac{V}{Z} = \frac{V_m \sin \omega t}{\sqrt{R^2 + X_L^2}\ \angle \tan^{-1}\frac{X_L}{R}}$$

$$i = \frac{V_m}{\sqrt{R^2 + X_L^2}} \sin\left(\omega t - \tan^{-1}\frac{X_L}{R}\right) \qquad \ldots(4.49)$$

or $$i = I_m \sin(\omega t - \phi)$$

where $$I_m = \text{Maximum value of current} = \frac{V_m}{\sqrt{R^2 + X_L^2}}$$

and ϕ = Angle between voltage and current (power factor angle)

$$= \tan^{-1} X_L/R = \tan^{-1} \omega L/R \qquad \ldots(4.50)$$

The impedance triangle and voltage current waveforms are shown in Fig. 4.23 a
4.24.

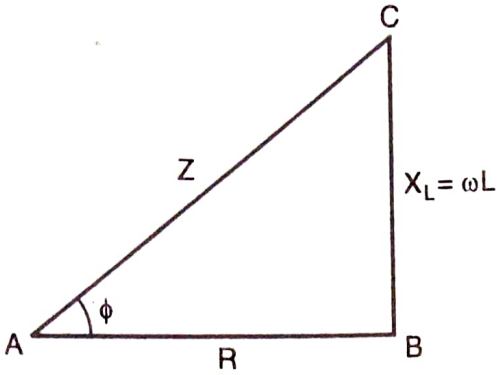


Fig. 4.23. Impedance Triangle.

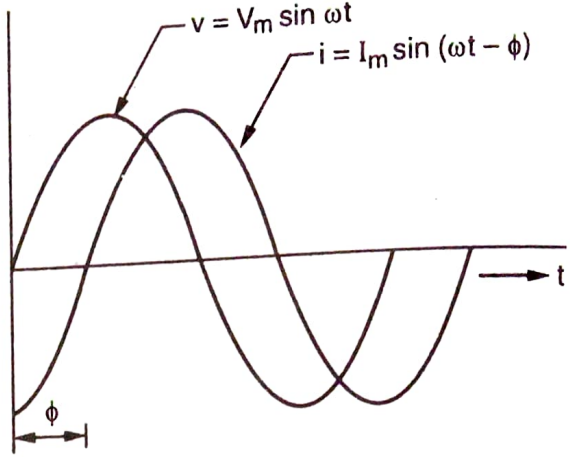


Fig. 4.24. Voltage, current Waveform.

Power :

Active power. The mean power consumed by the circuit is given by the product of V and *that component of the current I which is in phase with V.*

So, $$P = V I \cos \phi = \text{r.m.s. voltage} \times \text{r.m.s. current} \times \cos \phi \quad \ldots(4.51)$$

The term 'cos ϕ' is called the *power factor of the circuit.*

It should be noted that power consumed is due to ohmic resistance only because pure inductance does not consume any power.

$$P = VI \cos \phi$$

Now,

$$= VI \times (R/Z) \quad = (V/Z) \times I.R$$

$$= I^2 R \quad \text{or } P = I^2 R \text{ watt} \qquad (\because \cos \phi = R/Z)$$

Power factor. Power factor angle (angle between voltage and current)

$$\phi = \tan^{-1} \frac{X_L}{R} = \tan^{-1} \frac{\omega L}{R} \qquad ...(4.52)$$

ϕ is in between 0° and 90°

So, p.f. = cos ϕ is between unity and 0 (lagging).

4.6.6. A.C. through resistive capacitive circuit. Fig. 4.25 shows the circuit diagram having one resistor and pure capacitor connected in series. Again the applied voltage v is given by

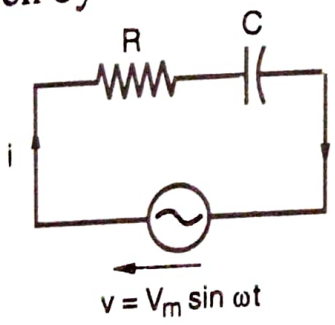


Fig. 4.25. Circuit Diagram.

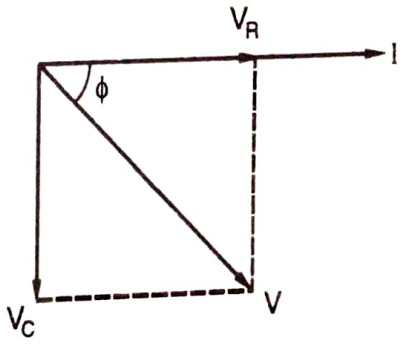


Fig. 4.26. Phasor Diagram

$$v = V_m \sin \omega t$$

Capacitive reactance, $$X_c = \frac{1}{\omega C}$$

Impedance of the circuit, $$Z = R - j X_c = R - j \frac{1}{\omega C} \qquad ...(4.53)$$

In polar form, $$Z = \sqrt{R^2 + X_c^2} \angle - \tan^{-1} \frac{X_C}{R}$$

In polar

The resultant current, i

$$i = \frac{V}{Z} = \frac{V_m \sin \omega t}{\sqrt{R^2 + X_c^2} \angle - \tan^{-1} X_C/R}$$

$$= \frac{V_m}{\sqrt{R^2 + X_c^2}} \sin (\omega t + \tan^{-1} X_C/R) \qquad ...(4.54)$$

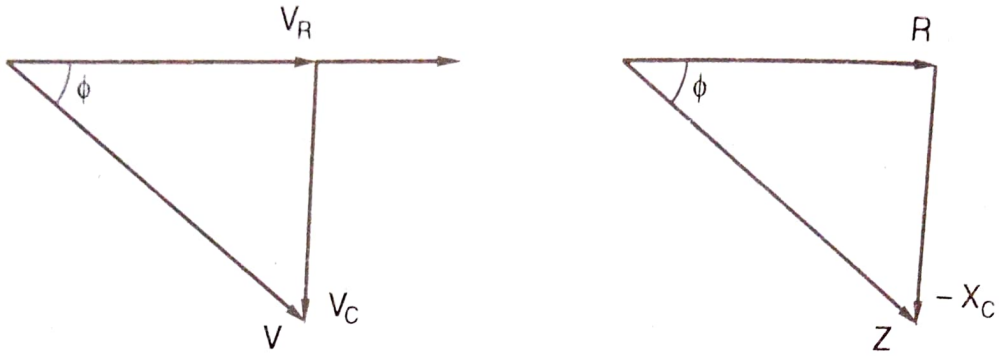
or $$i = I_m \sin (\omega t - \phi)$$

where I_m = Maximum value of current $= \dfrac{V_m}{\sqrt{R^2 + X_c^2}}$

ϕ = Angle between voltage and current (Power factor angle)

$$= - \tan^{-1} X_C/R = - \tan^{-1} \frac{1}{\omega CR} \qquad ...(4.55)$$

Voltage and impedance triangles are shown in Fig. 4.27.



(a) Voltage Triangle (b) Impedance Triangle

Fig. 4.27.

Here also we can define active component of current ($I \cos \phi$) and reactive component of current ($I \sin \phi$).

Average power consumed by the pure capacitor is zero, so the power consumed in the circuit is given by the active power.

$$P = I^2R = VI \cos \phi \text{ watts.} \qquad \ldots(4.56)$$

Power factor. Angle between voltage and current,

$$\phi = -\tan^{-1} \frac{X_C}{R} \qquad \text{(Power factor angle)}$$

Negative sign shows that the current is leading voltage by angle ϕ.

Power factor angle is in between 0° and 90°.

Power factor = $\cos \phi$

= between unity and 0 leading

Fig. 4.28 shows the voltage and current waveforms. Current is leading by angle ϕ.

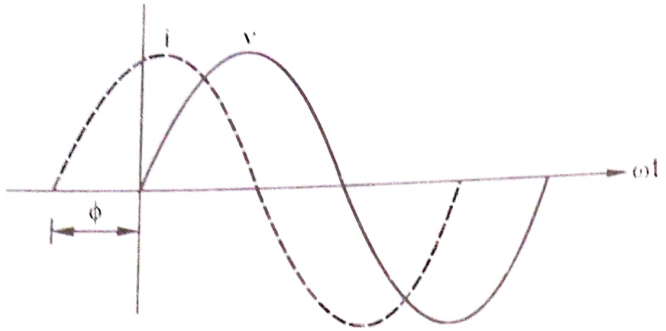


Fig. 4.28. Waveforms.

4.6.7. A.C. through series R-L-C circuit. Consider a R-L-C series containing resistance of $R\ \Omega$. L henry and C farad are connected across AC voltage of V volts (rms value) and frequency or f cycles/sec, as shown in Fig. 4.29.

The applied voltage is $v = V_m \sin \omega t$

Inductive reactance, $X_L = \omega L$

Capacitive reactance, $X_C = \dfrac{1}{\omega C}$

Impedance of the circuit,

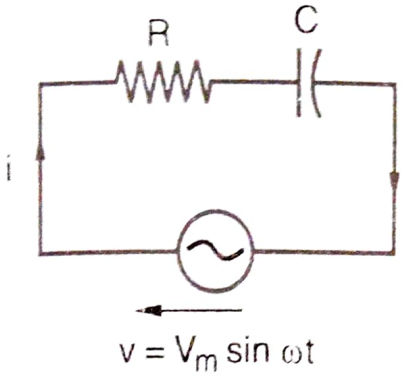


Fig. 4.29. Circuit Diagram.

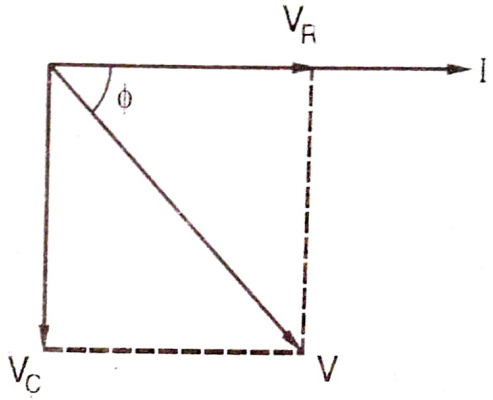


Fig. 4.30. Phasor Diagram.

$$Z = R + j\,(X_L - X_C) = R + j\left(\omega L - \frac{1}{\omega C}\right) \qquad ...(4.57)$$

In polar form, $$Z = \sqrt{R^2 + X^2}\ \angle \tan^{-1}\frac{X}{R}$$

where $$X = X_L - X_C$$

The resultant current i in the circuit,

$$i = \frac{v}{Z} = \frac{V_m \sin \omega t}{\sqrt{R^2 + X^2}\ \angle \tan^{-1}\frac{X}{R}}$$

$$= \frac{V_m}{\sqrt{R^2 + X^2}} \sin\left(\omega t - \tan^{-1}\frac{X}{R}\right). \qquad ...(4.58)$$

Now the

Now there are two case :

(*i*) $X_L > X_C$ or $V_L > V_C$

where V_L = Voltage across the inductor $= I X_L$

V_C = Voltage across the capacitor $= I X_C$

In this case nature of the circuit is inductive. In this case current I lags applied voltage v by angle ϕ.

$$\phi = \tan^{-1} \frac{X_L - X_C}{R} \quad \text{angle of lag}$$

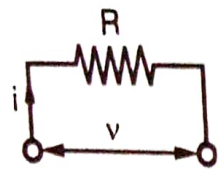
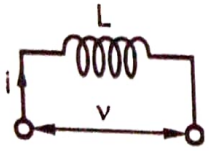
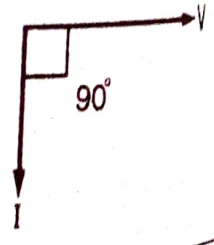
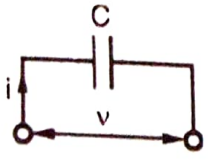
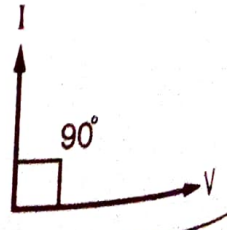
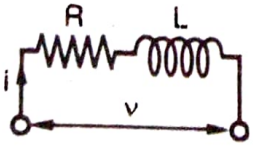
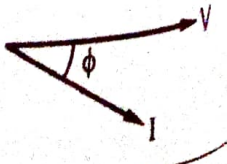
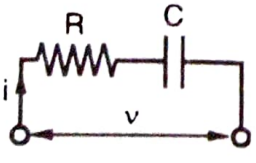
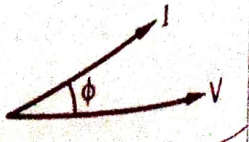
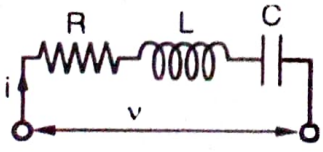
(*ii*) $X_C > X_L$ or $V_C > V_L$

In this case nature of the circuit is capacitive. Current i leads the applied voltage v by angle ϕ.

$$\phi = \tan^{-1} \frac{X_C - X_L}{R} \quad \text{angle of lead.}$$

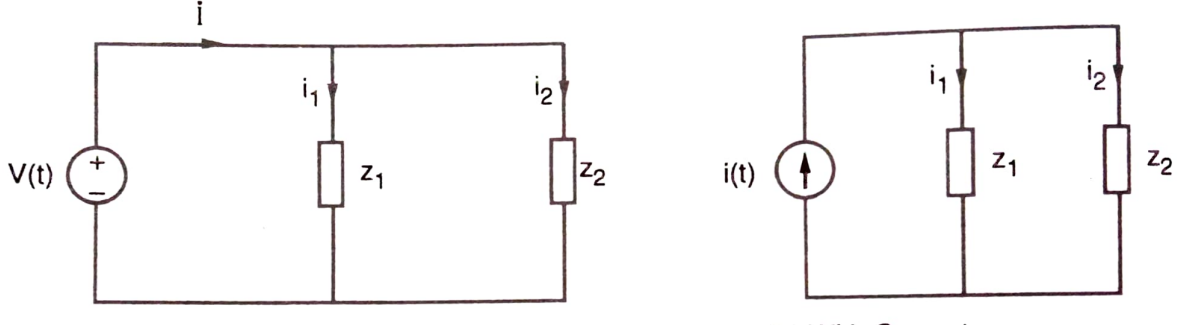
Table 4.1 shows the summary of results obtained in Article 4.6.

Supply voltage $v = V_m \sin \omega t$

Circuit	Impedance Z (Ω)	Current i (Amp)	Power factor angle, ϕ	Phasor diagram
 $v = V_m \sin \omega t$	$Z = R\ \Omega$	$i = \frac{V_m}{R} \sin \omega t$	$\phi = 0°$	
	$Z = j X_L$ $= j\omega L\ \Omega$	$i = \frac{V_m \sin (\omega t - 90°)}{X_L}$	$\phi = 90°$ (lagging)	
	$Z = -j X_C\ \Omega$ $= -j/\omega C$	$i = \frac{V_m}{X_C} \sin (\omega t + 90°)$	$\phi = 90°$ (leading)	
	$Z = R + j X_L$	$i = \frac{V_m}{\lvert Z \rvert} \sin (\omega t - \phi)$	$\phi = \tan^{-1} \frac{X_L}{R}$ (lagging) $0 < \phi < 90°$ lag	
	$Z = R - j X_C$	$i = \frac{V_m}{\lvert Z \rvert} \sin (\omega t + \phi)$	$\phi = \tan^{-1} \frac{X_C}{R}$ (leading) $0° < \phi < 90°$ lead	
	$Z = R + j (X_L - X_C)$	$i = \frac{V_m}{\lvert \dot{Z} \rvert} \sin (\omega t \pm \phi)$	$\phi = \tan^{-1} \frac{X_L - X_C}{R}$	

4.7. PARALLEL A.C. CIRCUIT

When two or more branches (or impedances) are connected in parallel across the source then it forms the parallel A.C. circuit. Fig. 4.31 shows such circuits with voltage and current source.



(a) With voltage source (b) With Current source

Fig. 4.31. Parallel A.C. Circuit.

Voltage acorss all the parallel branches remains same, while current is distributed.

By KCL at node, $i = i_1 + i_2$

According to current division rule :

$$i_1 = i \frac{Z_2}{Z_1 + Z_2} \quad \text{and} \quad i_2 = i \frac{Z_1}{Z_1 + Z_2}$$

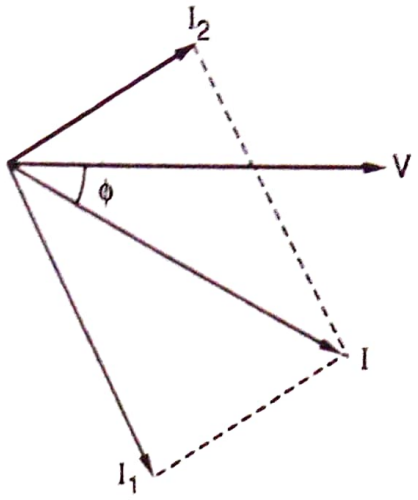


Fig. 4.32. Phasor diagram.

Fig. 4.32 shows the phasor diagram of parallel circuit. Take voltage as reference. The resultant current is the vector summation of i_1 and i_2.

Here it is assumed that I_2 is leading while I_1 is lagging behind the voltage. It means it is assumed that Z_1 is inductive and Z_2 is capacitive. ϕ is the p.f. angle.

Equivalent impedance. The equivalent impedance of parallel impedances $Z_1, Z_2, \ldots Z_n$ is given by Z, then

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} + \ldots + \frac{1}{Z_n} \qquad \ldots(4.59)$$

or

$$Y = Y_1 + Y_2 + \ldots + Y_n \qquad \ldots(4.60)$$

where Y is the admittance reciprocal of impedance. The admittance is given by

$$Y = G + jB \text{ in siemens or mho}$$

where G is the conductance and B is the susceptance. The equivalent impedance for circuit of Fig. 4.32

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

or

$$Z = \frac{Z_1 Z_2}{Z_1 + Z_2} \qquad \ldots(4.61)$$

Different parallel circuits :

(*i*) **Parallel *R*, *L*, *C* circuit.** In Fig. 4.33, X_L is the inductive reactance and X_C is the capacitive reactance, then

$$i_1 = \frac{v}{R}$$

$$i_2 = \frac{v}{j X_L} = \frac{v}{X_L} \angle -90^\circ \quad \text{(Current is lagging)}$$

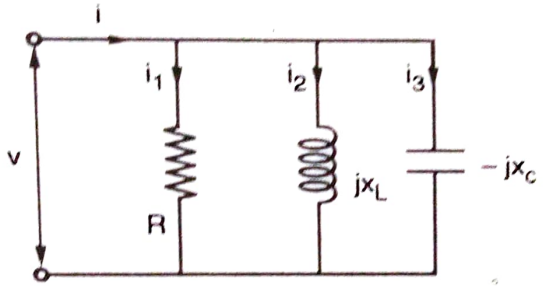


Fig. 4.33.

and $$i_3 = \frac{v}{-jX_c} = \frac{v}{X_c} \angle 90^\circ \text{ (Current is leading).}$$

Also, $$i = i_1 + i_2 + i_3 \quad \text{by KCL}$$

Equivalent Admittance. Let the equivalent impedance is Z, then,

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{-jX_C} = \frac{1}{R} + j\left(\frac{1}{X_C} - \frac{1}{X_L}\right)$$

Now admittance $$Y = \frac{1}{R} + j\left(\frac{1}{X_C} - \frac{1}{X_L}\right) = G + jB$$

where $$G = \frac{1}{R} \text{ and } B = \frac{1}{X_C} - \frac{1}{X_L}$$

(*ii*) Parallel combination of inductive and capacitive impedances.

Impedance of inductive branch : $Z_1 = R_1 + jX_L$

Impedance of capacitive branch : $Z_2 = R_2 - jX_C$

Now $$i_1 = \frac{v}{Z_1} = \frac{v}{R_1 + jX_L}$$

and $$i_2 = \frac{v}{Z_2} = \frac{v}{R_2 - jX_C}$$

Equivalent impedance : Y $$= Y_1 + Y_2 = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$= \frac{1}{R_1 + j\,X_L} + \frac{1}{R_2 - j\,X_C}$$

This can be converted into conductance and susceptance form.

In Fig. 4.34 i_1 is the lagging current and it lags by angle ϕ_1 ($= \tan^{-1} X_L/R$) behind the voltage. i_2 is the leading current and it leads the voltage by angle ϕ_2 ($= \tan^{-1} X_C/R$). The phasor diagram of this circuit is same as Fig. 4.32.

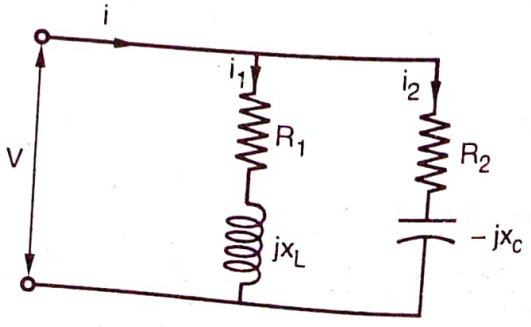


Fig. 4.34.